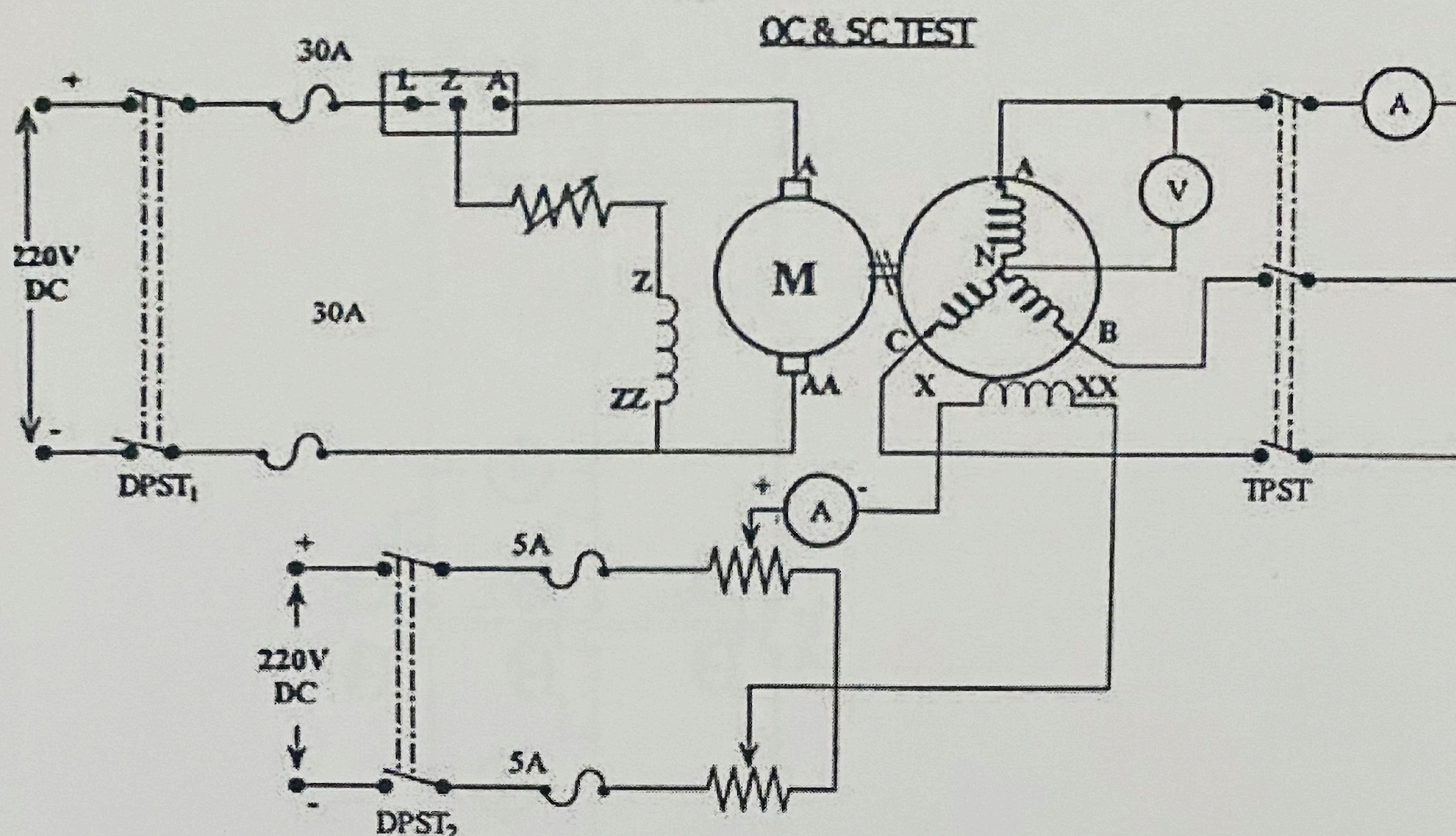




b. Zero Power Factor Characteristics (ZPFC)

Make the connections as shown in diagram.

Precautions- Initial settings:

- i) TPST in open position
- ii) DPST₁ and DPST₂ in open position
- iii) Motor field rheostat in minimum position
- iv) Potential divider in minimum voltage position

Switch on the DC supply to the DC motor by closing the switch DPST₁. Start the DC shunt motor using 3-point starter. Increase the resistance of dc motor field rheostat and drive the alternator at rated speed. Now, switch on the dc supply to the alternator field by closing the switch DPST₂; and vary the potential divider so that the generated voltage is nearly equal to rated value. Close the switch TPST₁ and note down the 3 phase supply voltage. Adjust the potential divider and make the generator terminal voltage equal to the 3 phase supply voltage. Now, the lamps will flicker uniformly (All the lamps become dim or bright at a time). If the lamps are not flickering uniformly (phase sequence is wrong), then interchange the two terminals of the 3 phase supply voltage after opening the switch TPST₁. If the flickering is so fast, the motor field rheostat is adjusted very slightly so that the frequency of flickering is convenient and the synchronization switch is closed at the middle of the dark period. Synchronization is over. After synchronization, adjust the motor field rheostat and make the wattmeter reading equal to zero. Now, increase the synchronous generator field current by varying the potential divider so that the armature current reaches rated value.